Requirement Document: N8N Hands-On Beginner Course

Topic 5: Expressions & Variables

1. Document Control

Field	Value
Document Title	Requirement Document: Topic 5 - Expressions & Variables
Course	N8N - Hands On Beginner Course
Target Audience	Trainee / Employee
Version	1.0
Status	Approved
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2. Objectives & Scope

2.1 Course Objective

The main objective of this learning module is to teach trainees and employees how to use n8n expressions to reference data dynamically, combine strings, change text casing, and format dates using the built-in Luxon date functions.

2.2 Learning Scope

In Scope:

- Referencing variables dynamically from parent nodes using n8n's `{{ ... }}` expression syntax.
- Combining multiple string fields together (concatenation) with custom separators.
- Performing basic text casing operations such as converting text to uppercase (using `.toUpperCase()`).
- Working with dates by parsing ISO-formatted date strings using n8n's built-in `$dateTime` helper.
- Formatting dates into human-readable representations using the `.toFormat()` syntax.

- Accessing the current date/time dynamically using the global `$now` variable.

Out of Scope:

- Advanced JavaScript string regex matching or complex string splitting/slicing logic.
- Complex date calculations like business days difference or calendar conversions.
- Installing external NPM modules or configuring custom scripting runtimes.

3. Target Audience Requirements (Pre-requisites)

To successfully complete this hands-on exercise, trainees/employees must have:

- Completed Topic 1 through Topic 4.
- Familiarity with basic JavaScript syntax (variables, operations, method calls).
- Access to an active n8n editor session with expressions capabilities.

4. Functional Requirements

Requirement ID	Section	Requirement Description	Priority
REQ-001	Input Setup	The workflow shall start with a node configured to output three mock data fields: `firstName` (all lowercase), `lastName` (all lowercase), and `birthDate` (an ISO date-time string).	**High**
REQ-002	Concatenation & Case	The workflow shall combine the `firstName` and `lastName` fields separated by a space and convert the result entirely to uppercase, mapped to `fullNameUpperCase`.	**High**
REQ-003	Date Formatting	The workflow shall parse the input `birthDate` ISO string and format it into `MMMM dd, yyyy` (e.g. "April 12, 1998"), mapped to `formattedBirthDate`.	**High**
REQ-004	Current Date	The workflow shall dynamically retrieve the current date/time using n8n's date variables and format it as `yyyy-MM-dd`, mapped to `processedDate`.	**High**

Requirement ID	Section	Requirement Description	Priority
REQ-005	Output Structure	The final workflow execution shall output a single JSON object containing exactly these fields: `fullNameUpperCase`, `formattedBirthDate`, and `processedDate`.	**High**

5. Technical Requirements

ID	Technical Requirement	Details
TR-001	Date Library (Luxon)	The system must support the Luxon-based datetime helper objects (`$dateTime` and `$now`) within expressions to format date strings.
TR-002	Expression Evaluation	The expressions engine must evaluate JavaScript string operations like `.toUpperCase()` and standard string concatenation (using `+` operator or template literals) in real time.

6. Stakeholders and Personas

Persona	Role	Need
Trainee / Employee	Student	Needs to learn how to clean, format, and prepare text and date fields dynamically when passing payloads between systems.
Instructor / Reviewer	Evaluation	Needs to confirm the student can write functional expressions and format values dynamically without hardcoding static values.

7. Assumptions & Constraints

- **Assumption 1 (ASM-001):** The trainee's n8n environment has standard configuration options enabled to resolve timezone expressions relative to UTC or the system locale.
- **Constraint 1 (CON-001):** Date formatting options are constrained to the supported tokens in Luxon's `toFormat` syntax.